SIMULATION BASED ROOM SECURITY AND HOME AUTOMATION SYSTEM

1st Semester ECE Mini Project

GROUP MEMBERS:

- PRATIK TANTY: BTECH/10211/25
- PIYUSH PRANJAL: BTECH/10212/25
- P SAMPAT GANESH: BTECH/10221/25
- KSHITIJ KUMAR: BTECH/10224/25

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
BIRLA INSTITUTE OF TECHNOLOGY, MESRA

PROJECT OBJECTIVE

GOAL:

To design a dual-purpose digital logic circuit that provides password-based security and automatic control of appliances.

KEY FEATURES

DIGITAL LOCK

Compares user input against a stored password for authentication, using logic gates

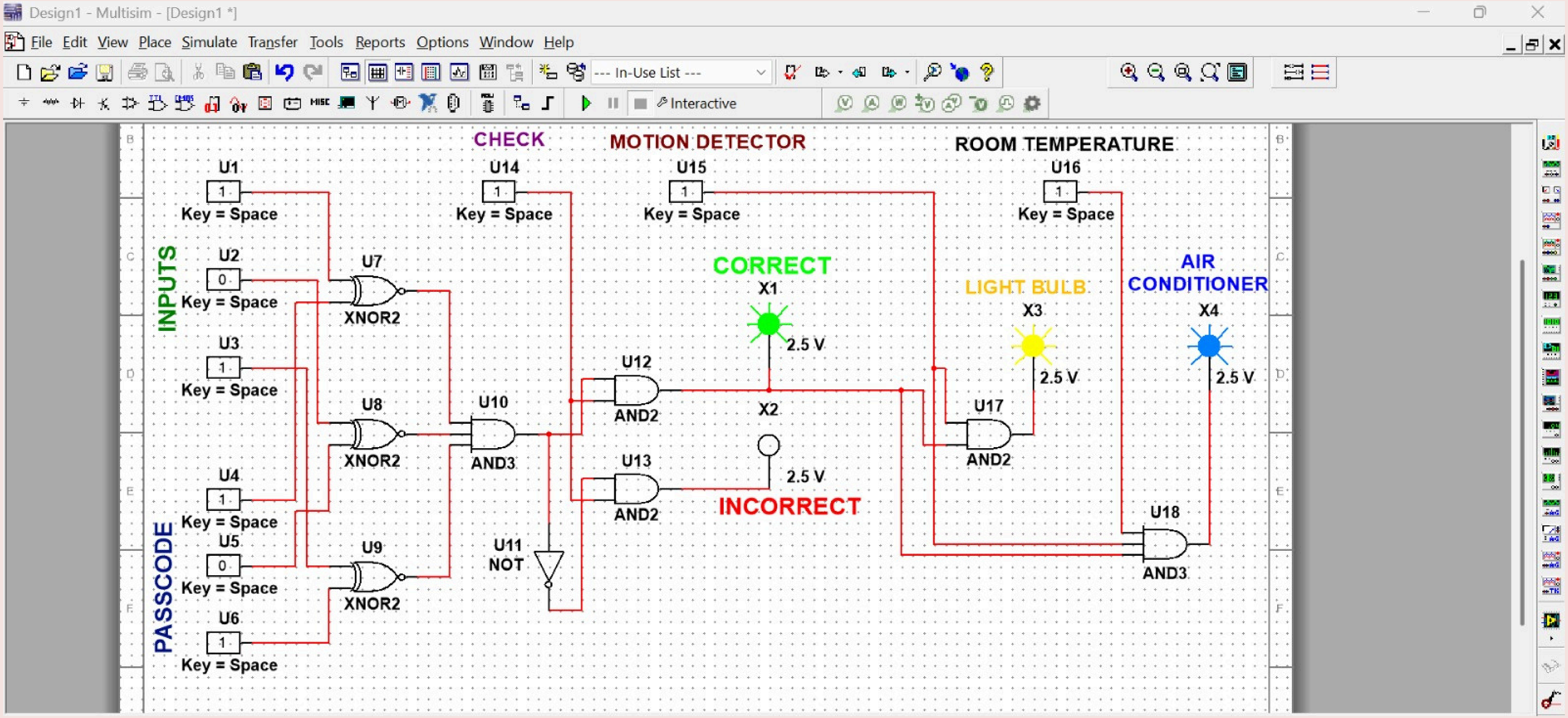
INTRUDER ALERT

Detects motion using motion detectors after authorizing access when correct password is entered

SMART AUTOMATION

Controls lights and Air Conditioning based on sensors only when user is authorized

CIRCUIT DIAGRAM



CIRCUIT DIAGRAM DESIGNED AND SIMULATED USING NI MULTISIM 14.1

COMPONENT DETAILS AND SPECIFICATIONS

01 **LOGIC GATES**

XNOR (comparison), AND, OR, NOT (control logic)

02 **INDICATORS**

Probe LEDs: Green (Correct), Red (Incorrect), Yellow (Light), Blue (AC)

03 **INPUTS**

Interactive Digital Keys (Space Key)

REAL LIFE APPLICATIONS

HOME SECURITY

Entry systems for houses & apartments

SERVER ROOMS

Where cooling & access control matter

HOTELS

Auto-lights off when guests leave

WAREHOUSE SECURITY

Lighting activates only in occupied sections to save energy

THANK YOU